

Skill 8:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define MAX_INPUT 500

/* Built-in command functions */

void builtin_cd(char *arg)
{
    if (arg == NULL)
    {
        printf("cd: missing argument\n");
        return;
    }

    if (chdir(arg) != 0)
    {
        perror("cd");
    }
}

void builtin_pwd()
{
    char cwd[500];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("%s\n", cwd);
}

void builtin_echo(char *arg)
{
    if (arg != NULL)
        printf("%s\n", arg);
}
```

```

void builtin_pwd()
{
    char cwd[500];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("%s\n", cwd);
}

void builtin_echo(char *arg)
{
    if (arg != NULL)
        printf("%s\n", arg);
}

/* Dispatch table */

typedef void (*builtin_func)(char *);

typedef struct
{
    char *name;
    builtin_func function;
} Builtin;

/* Built-in command dispatch table */

Builtin builtins[] =
{
    {"cd", builtin_cd},
    {"pwd", NULL},
    {"echo", builtin_echo}
};

int builtin_count = 3;

/* Find environment variable */

```

```

        while ((input[i] >= 'A' && input[i] <= 'Z') ||
               (input[i] >= 'a' && input[i] <= 'z') ||
               (input[i] >= '0' && input[i] <= '9') ||
               input[i] == '_')
        {
            variable[k++] = input[i++];
        }

        variable[k] = '\0';

        char *value = get_variable_value(variable);

        if (value != NULL)
        {
            /* Expand variable value */

            for (int x = 0; value[x] != '\0'; x++)
                output[j++] = value[x];
        }
        else
        {
            /* Undefined variable */

            printf("Undefined variable: %s\n", variable);
        }
    }
    else
    {
        output[j++] = input[i++];
    }
}

output[j] = '\0';
}

/* Execute built-in command */

```

```

strcpy(input, "$USER lives in $HOME");
expand_variables(input, expanded);

printf("\nNested variable test:\n");
printf("Original: %s\n", input);
printf("Expanded: %s\n", expanded);

/* Built-in command tests */

printf("\n=====\n");
printf("BUILT-IN COMMAND DISPATCH\n");
printf("=====\n");

printf("\nTesting echo:\n");
execute_builtin("echo", "Hello from built-in command");

printf("\nTesting pwd:\n");
execute_builtin("pwd", NULL);

printf("\nTesting cd:\n");
execute_builtin("cd", "/tmp");

printf("Current directory after cd:\n");
execute_builtin("pwd", NULL);

printf("\nTesting invalid command:\n");
execute_builtin("unknown", NULL);

printf("\n=====\n");
printf("PRACTICAL COMPLETED\n");
printf("=====\n");

return 0;
}

```

```
bhanu@DESKTOP-DE32HQ8:~$ cd ~
mkdir -p ~/shell_expansion
cd ~/shell_expansion
bhanu@DESKTOP-DE32HQ8:~/shell_expansion$ cd ~
mkdir -p ~/shell_expansion
cd ~/shell_expansion
bhanu@DESKTOP-DE32HQ8:~/shell_expansion$ pwd
/home/bhanu/shell_expansion
bhanu@DESKTOP-DE32HQ8:~/shell_expansion$ nano shell_expansion.c
bhanu@DESKTOP-DE32HQ8:~/shell_expansion$ gcc shell_expansion.c -o shell_expansion
bhanu@DESKTOP-DE32HQ8:~/shell_expansion$ ls
shell_expansion  shell_expansion.c
bhanu@DESKTOP-DE32HQ8:~/shell_expansion$ ./shell_expansion
=====
VARIABLE EXPANSION AND BUILT-INS
=====

Original: Hello $USER
Expanded: Hello bhanu

Original: Home directory: $HOME
Expanded: Home directory: /home/bhanu
Undefined variable: $UNDEFINED_VARIABLE
Original: Value: $UNDEFINED_VARIABLE
Expanded: Value:

Nested variable test:
Original: $USER lives in $HOME
Expanded: bhanu lives in /home/bhanu

=====
BUILT-IN COMMAND DISPATCH
=====

Testing echo:
Hello from built-in command

Testing pwd:
/home/bhanu/shell_expansion

Testing cd:
```

Expanded: bhanu lives in /home/bhanu

=====
BUILT-IN COMMAND DISPATCH
=====

Testing echo:
Hello from built-in command

Testing pwd:
/home/bhanu/shell_expansion

Testing cd:
Current directory after cd:
/tmp

Testing invalid command:
Invalid command: unknown

=====
PRACTICAL COMPLETED
=====

bhanu@DESKTOP-DE32HQ8:~/shell_expansion\$ |